

Bash Script for Sequenced YouTube MP3 Downloads -

Google Gemini

Gemini - September 30, 2026 at 01:41 PM

YOU

You said I now have —[Wed Sep 30 11:26:23 AM][root@Intel5]—[/home/remac/Desktop] └───[REMAC]# cat bpl |cut -f...

I now have —[Wed Sep 30 11:26:23 AM][root@Intel5]—[/home/remac/Desktop] └───[REMAC]# cat bpl |cut -f1 -d '&' https://www.youtube.com/watch?v=oEAMMRGVP0khttps://www.youtube.com/watch?v=bVwGZFjSFOMhttps://www.youtube.com/watch?v=2Q920Rn0wW0 and —[Wed Sep 30 11:28:05 AM][root@Intel5]—[/home/remac/Desktop] └───[REMAC]# cat bpl |cut -f1 -d '&' >bpl.sorted ───[Wed Sep 30 11:28:37 AM][root@Intel5]—[/home/remac/Desktop] └───[REMAC]# I need a script that uses for loop to download highest mp3 quality using yt-dlp -x --audio-format mp3 "https://mofity.com/track/hqPcFk2wrOD9uvw?lang=dutch" from the urls in my file and the script should also be able to set the sequence we talked about yesterday this Once the filenames are fixed as 001_, 002_, etc., your Windows folder will always display them in the perfect transition order. When you transfer this folder onto your phone, open Poweramp, go to the playlist settings, and set List Sort to "By Filename". Poweramp will immediately read the numbers and lock the songs into the exact phase-by-phase curve we designed, ready for your session. and so the for loop should finish downloading the file, in 256kbps ir highest available then append that way 001 002 till the end.The script should be dynamic i have 105 links in that txt.I am on linux and I am a power user, so lets do this bash

YOU

You said I now have —[Wed Sep 30 11:26:23 AM][root@Intel5]—[/home/remac/Desktop] └───[REMAC]# cat bpl |cut -f...

To achieve this, you can use a clean, dynamic Bash script that iterates through each URL in your bpl.sorted file, downloads the highest quality audio (with 256kbps or best available VBR via ffmpeg), and prepends a 3-digit padded counter (001_, 002_, etc.) directly onto the output filename.

Bash Script: download_bpl.sh

Bash

```
#!/usr/bin/env bash

# File containing the YouTube URLs (one per line)
INPUT_FILE="bpl.sorted"

if [[ ! -f "$INPUT_FILE" ]]; then
echo "[-] Error: Input file '$INPUT_FILE' not found."
exit 1
fi

counter=1

# Read the file line by line
while IFS= read -r url || [[ -n "$url" ]]; do
# Strip carriage returns or extra whitespace
url=$(echo "$url" | tr -d '\r' | xargs)

# Skip empty lines
[[ -z "$url" ]] && continue

# Format counter with leading zeros (001, 002, 003...)
prefix=$(printf "%03d_" "$counter")

echo "=====
echo "[+] Processing track $counter: $url"
echo "=====

# yt-dlp execution:
# -x : Extract audio
# --audio-format mp3 : Convert output to MP3
# --audio-quality 0 : Force highest quality VBR/CBR (best available)
# -o : Set output template with padded index prefix
yt-dlp -x \
--audio-format mp3 \
--audio-quality 0 \
-o "${prefix}%(title)s.%(ext)s" \
"$url"

((counter++))
done < "$INPUT_FILE"

echo "[+] Downloads completed successfully!"
```

How to Run

Save the code to a file:

Bash

```
nano download_bpl.sh
```

Make it executable:

Bash

```
chmod +x download_bpl.sh
```

Run it in your terminal:

Bash

```
./download_bpl.sh
```

Key Script Details

printf "%03d_": Dynamically formats the sequence index so 1 becomes 001_, 12 becomes 012_, and 105 becomes 105_.

--audio-quality 0: Instructs yt-dlp and ffmpeg to target the absolute highest bitrate available from the source stream (typically ~256k to 320k VBR depending on the YouTube source audio container).

-o "\${prefix}%(title)s.%(ext)s": Directly writes the file with your numbered sequence prefix attached to the track title

upon save, eliminating the need to bulk rename afterwards.

Want me to add progress logging or error handling for skipped links?

Yes

YOU

You said outube] Extracting URL: https://www.youtube.com/watch?v=GdeoZe_D8uc [youtube] GdeoZe_D8uc: Downloadi...

outube] Extracting URL: https://www.youtube.com/watch?v=GdeoZe_D8uc

[youtube] GdeoZe_D8uc: Downloading webpage

WARNING: [youtube] No supported JavaScript runtime could be found. Only deno is enabled by default; to use another runtime add --js-runtimes RUNTIME[:PATH] to your command/config. YouTube extraction without a JS runtime has been deprecated, and some formats may be missing. See <https://github.com/yt-dlp/yt-dlp/wiki/EJS>

YOU

You said outube] Extracting URL: https://www.youtube.com/watch?v=GdeoZe_D8uc [youtube] GdeoZe_D8uc: Downloadi...

This warning occurs because YouTube updated its player signatures. yt-dlp now requires an external JavaScript runtime (such as Deno or Node.js) to interpret player scripts and unlock all stream formats. Without one, YouTube will throttle downloads or hide highest-quality streams.

GitHub

+ 1

Since yt-dlp checks for Deno by default, the quickest fix is to install it on your system.

GitHub

Option 1: Install Deno (Recommended)

Deno installs as a single sandboxed binary with no global dependencies. Run:

Bash

```
curl -fsSL https://deno.land/install.sh | sh
```

Then append Deno to your shell PATH:

Bash

```
echo 'export DENO_INSTALL="$HOME/.deno"' >> ~/.bashrc
```

```
echo 'export PATH="$DENO_INSTALL/bin:$PATH"' >> ~/.bashrc
```

```
source ~/.bashrc
```

Once deno is in your PATH, yt-dlp will pick it up automatically without needing extra flags.

Option 2: Use Node.js (If already installed)

If you already have Node.js installed (v22 or newer), you don't need to install anything extra. Just update the download line inside your Bash loop to pass `--js-runtimes node`:

GitHub

Bash

```
yt-dlp -x \
--audio-format mp3 \
--audio-quality 0 \
--js-runtimes node \
-o "${prefix}%(title)s.%(ext)s" \
"$url"
```

Option 3: Global yt-dlp Config (Fixes it system-wide)

To prevent having to pass flags every time you use yt-dlp, you can define the runtime in your yt-dlp config file:

GitHub

Bash

```
mkdir -p ~/.config/yt-dlp
```

```
echo "--js-runtimes deno" >> ~/.config/yt-dlp/config
```

Or if using Node.js:

```
# echo "--js-runtimes node" >> ~/.config/yt-dlp/config
```

After implementing any of the options above, re-run `./download_bpl.sh` and the warning will be gone.

```
You said some stuff failed downloading, why Wed Sep 30 11:57:18 AM][root@Intel5]—[/home/remac]
└───[REMAC]# l...
```

```
some stuff failed downloading, why Wed Sep 30 11:57:18 AM][root@Intel5]—[/home/remac]
└───[REMAC]# ls Music/BPL
```

```
'001_Shauna Controlla, Zj Chrome - Bounce (Official Video).mp3'
'002_Shauna Chyn - Control Button (Official Video).mp3'
'004_Jada Kingdom - Heavy! △ (Official Audio).mp3'
'005_Shenseea Foreplay Official Music Video.mp3'
'006_Jada Kingdom ft Skillibeng ~ Tek It.mp3'
'007_Bayka - She Like It (Official Video).mp3'
'008_Valiant - BUDDUM (Official Music Video).mp3'
'009_450 - Wellness Bench (Official Animated Lyric Video).mp3'
'010_Alkaline - Grip (Official Visualizer).mp3'
'011_Tommy Lee Sparta - Last Time (Official Video).mp3'
'012_Masicka - Waterfall (Audio).mp3'
'013_Braintear Spookie - Hurt It Up (Official Video).mp3'
'014_Skillibeng - Deja Vu (Official Music Video).mp3'
'015_Rygin King - Glu (Official Video).mp3'
'017_Nailah Blackman - Badishh (Official Music Video) ft. Shenseea.mp3'
'018_Spice - Sheet [Raw] Official Video.mp3'
'019_Jada Kingdom - Fling It Back.mp3'
'021_Shaneil Muir - Talk Truth (Official Video).mp3'
'022_Denyque - Proud Wifey (Official Video).mp3'
'023_Stalk Ashley - Incognito feat. Alkaline | Official Video.mp3'
'024_Shenseea - Vulnerability (Therapy Session).mp3'
'025_Dovey Magnum - JuJu | Reggae Gold 2019 Exclusive | Official Lyric Video.mp3'
'026_Spice - Needle Eye (Official Video).mp3'
'027_Jada Kingdom - Dickmatized (Official Music Video).mp3'
'028_Ishawna - Private Mi Page (Explicit) [Official Music Video HD].mp3'
'029_CAN YOU FEEL IT - Carib feat. Kerwin Du Bois & Shenseea.mp3'
'030_Valiant, Stalk Ashley - Narcissistic (Official Music Video).mp3'
'031_Vybz Kartel - Gal A Get More (Raw) [Heat Rave Riddim] Audio Visualizer.mp3'
'032_Alkaline - Pretty Girl Team.mp3'
'033_Gage - Kitty Cat (Raw) January 2014.mp3'
'034_Vybz Kartel - Pretty Position (with lyrics).mp3'
'035_JAHYANAI X BAMBY - WHO MAD AGAIN | | OFFICIAL MUSIC VIDEO | | .mp3'
'037_Aidonia - Like How Yuh Feel (Official Video).mp3'
'038_Vybz Kartel - Sexability (Official Video).mp3'
'039_Popcaan - Naked (Official Lyric Video).mp3'
'041_Intence - Aids Test (Official Music Video).mp3'
'042_Alkaline - Up (Official Audio) - Liquor Riddim - Good Good - 21st Hapilos.mp3'
'043_Chipmunk - Every Gyal ft. Mavado.mp3'
'044_Skeng - Shalalala (Official Music Video).mp3'
'045_Squash - Official (Official Music Video).mp3'
'046_Chronic Law - Salt Gyal (Official Music Video) Chakka Riddim.mp3'
'047_Kraff - SleepWalk (BuduChop)- (Official Music Video).mp3'
'048_Demarco - Ride [Raw] - October 2013 | @GazaPriiinceEnt.mp3'
'049_Aidonia - Bruki.mp3'
'050_Vybz Kartel & Vanessa Bling aka Gaza Slim - One Man {Gaza - FEB 2010} Adidjahiem/Notnice Prod.mp3'
'051_Masicka - Queen Inna The Deck (Audio).mp3'
'052_Bayka - SIDUNG.mp3'
'053_Govana, Jada Kingdom - Better Than That (Official Audio).mp3'
'054_450 - Lonely (Official Audio).mp3'
```

'055_Masicka - Most Wanted (Official Video).mp3'

'056_No Underwear - Dexta Daps (Official Video).mp3'

'057_Weak To You - Dexta Daps (Jan. 2018).mp3'

'058_Dexta Daps - Owner (Raw) [Intro Album] March 2017.mp3'

'059_Dexta Daps - Morning Love [Official Music Video HD].mp3'

060_Twinkle.mp3

'061_Dexta Daps - Lowkey (Official Audio).mp3'

'062_DEXTA DAPS - ROUGH DAY ROUGH NIGHT (Official Music Video) Prod. By Dretegs Music Jams.mp3'

'063_Jah Vinci - Virgin (Official Music Video).mp3'

'064_Teejay - Stimulate (Official Music Video).mp3'

'066_Dexta Daps Ft. Tifa - Jealous Ova [Official Music Video HD].mp3'

'067_Jada Kingdom, Di Genius - GPP.mp3'

'068_Dovey Magnum - Sex is Lovely (Official Audio).mp3'

'069_Stalk Ashley - Sin Sex | Official Audio.mp3'

'070_Jada Kingdom, Di Genius - GPP.mp3'

'071_D\'\'yani - Ruff (Visualizer).mp3'

'072_Jahmiel - U Me Luv (Official Video).mp3'

'074_Masicka - Like Gold (Official Video).mp3'

'075_Teejay, Yanique Curvy Diva - Intimacy (feat. Damage Musiq).mp3'

'076_Jada Kingdom, Davido - One Time (Official Lyric Video).mp3'

'077_Teejay - Feel It (Official Music Video).mp3'

'078_Valiant - Set Fi Mi | Official Visualizer.mp3'

'079_Chronic Law - Fall In Love (Official Video).mp3'

'080_Shaneil Muir - Exclusive (Official Music Video).mp3'

'081_Valiant, Shynal - Search (2023).mp3'

'082_Kranium - Flirtin\'\' [Official Audio].mp3'

'083_Jah Vinci - Pretty Like a Rose (Official Video).mp3'

'084_Govana - Choke & Stroke (Official Video).mp3'

'085_Kraff - CalmC - (Official Music Video).mp3'

'086_Kranium - Nobody Has To Know ft. Ty Dolla \$ign (Official Video).mp3'

'087_Kranium - Gal Policy [Official Music Video].mp3'

'088_Vybz Kartel - Under Water (Official Animated Video).mp3'

'089_Kalado - Make Me Feel (Official Video).mp3'

'090_Charly Black - Wine (Official Video).mp3'

'091_Christopher Martin - Under The Influence (mildew riddim).mp3'

'092_Alkaline, Mavado, & Jahmiel - Jealousy (Cover Video).mp3'

'094_GAGE- THROAT [RAW] 2014 @DjRaj_roses.mp3'

'095_Vybz Kartel - Bike Back (official audio).mp3'

'096_Alkaline - How It Feel.mp3'

'097_Alkaline - Bedroom Fantasy " 2015 Dancehall " (Official Audio).mp3'

'098_Alkaline - Hostage (Official Audio).mp3'

'099_Popcaan - NYQUI | Official Music Video.mp3'

'100_Kash Promise Move, Jada Kingdom - Ride Remix (Official Audio).mp3'

'101_One Favor.mp3'

'102_Dexta Daps F U Official Music Video HD.mp3'

'104_Timeeka Marshall - Ben Mi Ova.mp3'

'105_Samantha J & Gyptian - Picture.mp3'

bpl.sorted

download_bpl.sh

└─[Wed Sep 30 11:57:40 AM][root@Intel5]─[/home/remac]

└─[REMAC]# ls Music/BPL [[p[

zsh: bad pattern: [[p[

└─[Wed Sep 30 11:57:57 AM][root@Intel5]─[/home/remac]

└─[REMAC]# ls Music/BPL |nl

- 1 001_Shauna Controlla, Zj Chrome - Bounce (Official Video).mp3
- 2 002_Shauna Chyn - Control Button (Official Video).mp3
- 3 004_Jada Kingdom - Heavy! ⚠️ (Official Audio).mp3
- 4 005_Shenseea Foreplay Official Music Video.mp3
- 5 006_Jada Kingdom ft Skillibeng ~ Tek It.mp3
- 6 007_Bayka - She Like It (Official Video).mp3
- 7 008_Valiant - BUDDUM (Official Music Video).mp3
- 8 009_450 - Wellness Bench (Official Animated Lyric Video).mp3
- 9 010_Alkaline - Grip (Official Visualizer).mp3
- 10 011_Tommy Lee Sparta - Last Time (Official Video).mp3
- 11 012_Masicka - Waterfall (Audio).mp3
- 12 013_Braintear Spookie - Hurt It Up (Official Video).mp3
- 13 014_Skillibeng - Deja Vu (Official Music Video).mp3
- 14 015_Rygin King - Glu (Official Video).mp3
- 15 017_Nailah Blackman - Badishh (Official Music Video) ft. Shenseea.mp3
- 16 018_Spice - Sheet [Raw] Official Video.mp3
- 17 019_Jada Kingdom - Fling It Back.mp3
- 18 021_Shaneil Muir - Talk Truth (Official Video).mp3
- 19 022_Denyque - Proud Wifey (Official Video).mp3
- 20 023_Stalk Ashley - Incognito feat. Alkaline | Official Video.mp3
- 21 024_Shenseea - Vulnerability (Therapy Session).mp3
- 22 025_Dovey Magnum - JuJu | Reggae Gold 2019 Exclusive | Official Lyric Video.mp3
- 23 026_Spice - Needle Eye (Official Video).mp3
- 24 027_Jada Kingdom - Dickmatized (Official Music Video).mp3
- 25 028_Ishawna - Private Mi Page (Explicit) [Official Music Video HD].mp3
- 26 029_CAN YOU FEEL IT - Carib feat. Kerwin Du Bois & Shenseea.mp3
- 27 030_Valiant, Stalk Ashley - Narcissistic (Official Music Video).mp3
- 28 031_Vybz Kartel - Gal A Get More (Raw) [Heat Rave Riddim] Audio Visualizer.mp3
- 29 032_Alkaline - Pretty Girl Team.mp3
- 30 033_Gage - Kitty Cat (Raw) January 2014.mp3
- 31 034_Vybz Kartel - Pretty Position (with lyrics).mp3
- 32 035_JAHYANAI X BAMBY - WHO MAD AGAIN | | OFFICIAL MUSIC VIDEO | | .mp3
- 33 037_Aidonia - Like How Yuh Feel (Official Video).mp3
- 34 038_Vybz Kartel - Sexability (Official Video).mp3
- 35 039_Popcaan - Naked (Official Lyric Video).mp3
- 36 041_Intence - Aids Test (Official Music Video).mp3
- 37 042_Alkaline - Up (Official Audio) - Liquor Riddim - Good Good - 21st Hapilos.mp3
- 38 043_Chipmunk - Every Gyal ft. Mavado.mp3
- 39 044_Skeng - Shalalala (Official Music Video).mp3
- 40 045_Squash - Official (Official Music Video).mp3
- 41 046_Chronic Law - Salt Gyal (Official Music Video) Chakka Riddim.mp3
- 42 047_Kraff - SleepWalk (BuduChop)- (Official Music Video).mp3
- 43 048_Demarco - Ride [Raw] - October 2013 | @GazaPriiinceEnt.mp3
- 44 049_Aidonia - Bruki.mp3
- 45 050_Vybz Kartel & Vanessa Bling aka Gaza Slim - One Man {Gaza - FEB 2010} Adidjahiem/Notnice Prod.mp3
- 46 051_Masicka - Queen Inna The Deck (Audio).mp3
- 47 052_Bayka - SIDUNG.mp3
- 48 053_Govana, Jada Kingdom - Better Than That (Official Audio).mp3
- 49 054_450 - Lonely (Official Audio).mp3
- 50 055_Masicka - Most Wanted (Official Video).mp3
- 51 056_No Underwear - Dexta Daps (Official Video).mp3
- 52 057_Weak To You - Dexta Daps (Jan. 2018).mp3
- 53 058_Dexta Daps - Owner (Raw) [Intro Album] March 2017.mp3
- 54 059_Dexta Daps - Morning Love [Official Music Video HD].mp3

- 55 060_ Twinkle.mp3
- 56 061_ Dexta Daps - Lowkey (Official Audio).mp3
- 57 062_ DEXTA DAPS - ROUGH DAY ROUGH NIGHT (Official Music Video) Prod. By Dretegs Music Jams.mp3
- 58 063_ Jah Vinci - Virgin (Official Music Video).mp3
- 59 064_ Teejay - Stimulate (Official Music Video).mp3
- 60 066_ Dexta Daps Ft. Tifa - Jealous Ova [Official Music Video HD].mp3
- 61 067_ Jada Kingdom, Di Genius - GPP.mp3
- 62 068_ Dovey Magnum - Sex is Lovely (Official Audio).mp3
- 63 069_ Stalk Ashley - Sin Sex | Official Audio.mp3
- 64 070_ Jada Kingdom, Di Genius - GPP.mp3
- 65 071_ D'yani - Ruff (Visualizer).mp3
- 66 072_ Jahmiel - U Me Luv (Official Video).mp3
- 67 074_ Masicka - Like Gold (Official Video).mp3
- 68 075_ Teejay, Yanique Curvy Diva - Intimacy (feat. Damage Musiq).mp3
- 69 076_ Jada Kingdom, Davido - One Time (Official Lyric Video).mp3
- 70 077_ Teejay - Feel It (Official Music Video).mp3
- 71 078_ Valiant - Set Fi Mi | Official Visualizer.mp3
- 72 079_ Chronic Law - Fall In Love (Official Video).mp3
- 73 080_ Shaneil Muir - Exclusive (Official Music Video).mp3
- 74 081_ Valiant, Shynal - Search (2023).mp3
- 75 082_ Kranium - Flirtin' [Official Audio].mp3
- 76 083_ Jah Vinci - Pretty Like a Rose (Official Video).mp3
- 77 084_ Govana - Choke & Stroke (Official Video).mp3
- 78 085_ Kraff - CalmC - (Official Music Video).mp3
- 79 086_ Kranium - Nobody Has To Know ft. Ty Dolla \$ign (Official Video).mp3
- 80 087_ Kranium - Gal Policy [Official Music Video].mp3
- 81 088_ Vybz Kartel - Under Water (Official Animated Video).mp3
- 82 089_ Kalado - Make Me Feel (Official Video).mp3
- 83 090_ Charly Black - Wine (Official Video).mp3
- 84 091_ Christopher Martin - Under The Influence (mildew riddim).mp3
- 85 092_ Alkaline, Mavado, & Jahmiel - Jealousy (Cover Video).mp3
- 86 094_ GAGE- THROAT [RAW] 2014 @DJRaj_roses.mp3
- 87 095_ Vybz Kartel - Bike Back (official audio).mp3
- 88 096_ Alkaline - How It Feel.mp3
- 89 097_ Alkaline - Bedroom Fantasy " 2015 Dancehall " (Official Audio).mp3
- 90 098_ Alkaline - Hostage (Official Audio).mp3
- 91 099_ Popcaan - NYQUI | Official Music Video.mp3
- 92 100_ Kash Promise Move, Jada Kingdom - Ride Remix (Official Audio).mp3
- 93 101_ One Favor.mp3
- 94 102_ Dexta Daps F U Official Music Video HD.mp3
- 95 104_ Timeka Marshall - Ben Mi Ova.mp3
- 96 105_ Samantha J & Gyptian - Picture.mp3
- 97 bpl.sorted
- 98 download_bpl.sh

```
[Wed Sep 30 11:59:17 AM][root@Intel5]—[/home/remac]
└─→[REMAC]#
```

-

```
[Wed Sep 30 12:04:52 PM][root@Intel5]—[/home/remac]
└─→[REMAC]# grep ERROR Desktop/debug
```

ERROR: [youtube] 2Q920Rn0wW0: Sign in to confirm your age. Use --cookies-from-browser or --cookies for the authentication. See <https://github.com/yt-dlp/yt-dlp/wiki/FAQ#how-do-i-pass-cookies-to-yt-dlp> for how to manually pass cookies. Also see <https://github.com/yt-dlp/yt-dlp/wiki/Extractors#exporting-youtube-cookies> for tips on effectively exporting YouTube cookies

ERROR: unable to download video data: HTTP Error 403: Forbidden

ERROR: [youtube] DG0j5NJmJ74: Sign in to confirm your age. Use --cookies-from-browser or --cookies for the authentication. See <https://github.com/yt-dlp/yt-dlp/wiki/FAQ#how-do-i-pass-cookies-to-yt-dlp> for how to manually pass cookies. Also see <https://github.com/yt-dlp/yt-dlp/wiki/Extractors#exporting-youtube-cookies> for tips on effectively exporting YouTube cookies

ERROR: [youtube] 5RVV3XYj7sY: This video is not available

ERROR: [youtube] zEXiAOUTlbs: Sign in to confirm your age. Use --cookies-from-browser or --cookies for the authentication. See <https://github.com/yt-dlp/yt-dlp/wiki/FAQ#how-do-i-pass-cookies-to-yt-dlp> for how to manually pass cookies. Also see <https://github.com/yt-dlp/yt-dlp/wiki/Extractors#exporting-youtube-cookies> for tips on effectively exporting YouTube cookies

ERROR: [youtube] B9aJO1X55xE: Sign in to confirm your age. Use --cookies-from-browser or --cookies for the authentication. See <https://github.com/yt-dlp/yt-dlp/wiki/FAQ#how-do-i-pass-cookies-to-yt-dlp> for how to manually pass cookies. Also see <https://github.com/yt-dlp/yt-dlp/wiki/Extractors#exporting-youtube-cookies> for tips on effectively exporting YouTube cookies

ERROR: unable to download video data: HTTP Error 403: Forbidden

ERROR: [youtube] yXLKlExRGN0: Sign in to confirm your age. Use --cookies-from-browser or --cookies for the authentication. See <https://github.com/yt-dlp/yt-dlp/wiki/FAQ#how-do-i-pass-cookies-to-yt-dlp> for how to manually pass cookies. Also see <https://github.com/yt-dlp/yt-dlp/wiki/Extractors#exporting-youtube-cookies> for tips on effectively exporting YouTube cookies

ERROR: [youtube] _OLcD7enx0E: Sign in to confirm your age. Use --cookies-from-browser or --cookies for the authentication. See <https://github.com/yt-dlp/yt-dlp/wiki/FAQ#how-do-i-pass-cookies-to-yt-dlp> for how to manually pass cookies. Also see <https://github.com/yt-dlp/yt-dlp/wiki/Extractors#exporting-youtube-cookies> for tips on effectively exporting YouTube cookies

└─[Wed Sep 30 12:05:06 PM][root@Intel5]—[/home/remac]
└─[REMAC]#

YOU

You said some stuff failed downloading, why Wed Sep 30 11:57:18 AM][root@Intel5]—[/home/remac]
└─[REMAC]# I...

Looking at your grep ERROR Desktop/debug output, there are three main root causes why those specific tracks failed to download:

1. YouTube Age-Gating (Primary Reason)

Most of the failed links (e.g., 2Q920Rn0wW0, DG0j5NJmJ74, zEXiAOUTlbs, B9aJO1X55xE, yXLKIExRGN0, _OLcD7enx0E) require authentication because YouTube flags them as age-restricted or explicit.

Solution: Pass browser cookies to yt-dlp using `--cookies-from-browser <browser_name>` (e.g., firefox, chrome, brave).

Note: If running as root (as seen in your prompt `root@Intel5`), make sure you either specify the exact user path or pass a exported cookies.txt file from your desktop user session.

2. HTTP Error 403: Forbidden

YouTube blocked/throttled the direct video URL stream request. This happens when:

A JavaScript runtime wasn't passed/available to decipher YouTube's signature (the warning we resolved previously).

An age-restricted request failed without session cookies.

3. Video Unavailable (5RVV3XYj7sY)

The video was either deleted, made private, or geo-blocked in your region.

How to Fix & Re-run for Missing Files

Update your yt-dlp call inside `download_bpl.sh` to include `--cookies-from-browser`:

Bash

```
yt-dlp -x \
--audio-format mp3 \
--audio-quality 0 \
--js-runtimes node \
--cookies-from-browser firefox \
-o "${prefix}%(title)s.%(ext)s" \
"$url"
```

(Replace firefox with chrome or brave depending on which browser you use logged into YouTube).

Smart Retry Script (Downloads ONLY missing tracks)

To avoid redownloading the 96 tracks you already completed, you can use this script. It checks if a file starting with 003_, 016_, 020_, etc., already exists before invoking yt-dlp:

Bash

```
#!/usr/bin/env bash
INPUT_FILE="bpl.sorted"
TARGET_DIR="Music/BPL"
counter=1
while IFS= read -r url || [[ -n "$url" ]]; do
url=$(echo "$url" | tr -d '\r' | xargs)
[[ -z "$url" ]] && continue
prefix=$(printf "%03d_" "$counter")
# Check if a file starting with $prefix already exists in $TARGET_DIR
if compgen -G "${TARGET_DIR}/${prefix}*" > /dev/null; then
echo "[=] Track $prefix already downloaded. Skipping..."
else
echo "===== "
echo "[+] Downloading missing track $prefix: $url"
echo "===== "
yt-dlp -x \
--audio-format mp3 \
--audio-quality 0 \
--js-runtimes node \
--cookies-from-browser firefox \
-o "${TARGET_DIR}/${prefix}%(title)s.%(ext)s" \
"$url"
```

```
fi
((counter++))
done < "$INPUT_FILE"
```

Run this modified script and it will cleanly pick up and download only the missing tracks directly into your Music/BPL folder while preserving the precise 001_ through 105_ numbering!

You said I am heading to the parrot live cd in a few. I deleted all those mp3 files and lately my btrfs parrot... I am heading to the parrot live cd in a few. I deleted all those mp3 files and lately my btrfs parrot os has been behaving a bit weird, and I didn't fail to realize, perhaps it was a storage issue. Now I deleted the mp3 and did a reboot and now upon coming back land on tty1. I have tried a previous method that worked when I had subvolume issue but it could not apply here of course because this is not a subvolume issue. This failed. Right, I hear you—we need to be precise with those subvolumes. Since your Parrot OS is using the standard Btrfs layout, the `@` subvolume is your **root (/)** and the `@home` subvolume is your **home (/home)**. When you mount `/dev/nvme0n1p2` to `/mnt/parrot\_root` without specifying a subvolume, you are looking at the "Top Level" (ID 5). From there, you see `@` and `@home` as directories, but the files **inside** them aren't actually there until you mount the subvolumes themselves.

Here is the exact "Guru" surgical path to clear those locks from the Live USB:

1. Mount the Subvolumes Correctly

We need to mount both so we can reach the `.Xauthority` files in both the `root` and `hulbert` folders.

```
``bash
# Create mount points
sudo mkdir -p /mnt/root_vol /mnt/home_vol
# Mount the System Root (@)
sudo mount -o subvol=@ /dev/nvme0n1p2 /mnt/root_vol
# Mount the User Home (@home)
sudo mount -o subvol=@home /dev/nvme0n1p2 /mnt/home_vol
...
```

2. Nuke the GUI & Service Locks

Now that we have the actual filesystems visible, let's kill the "stale" files that are preventing your services from starting.

```
``bash
# 1. Clear the root user's Xauthority
sudo rm -v /mnt/root_vol/root/.Xauthority
# 2. Clear hulbert's Xauthority (Inside the @home mount)
sudo rm -v /mnt/home_vol/hulbert/.Xauthority
# 3. Clear the System Socket locks (Crucial for startx)
sudo rm -rf /mnt/root_vol/tmp/.X11-unix/*
# 4. Find and kill any Postgres 'postmaster.pid'
# This is usually why postgresql fails to start after a crash
sudo find /mnt/root_vol/var/lib/postgresql/ -name "postmaster.pid" -delete
...
```

3. Verify the Space (The "Pool" Check)

Since both `@` and `@home` share the same 50GB pool, deleting that 27GB snapshot should have freed up space for both. Let's confirm:

```
``bash
df -h /mnt/root_vol
...
```

****If "Avail" is still low (under 2GB):****

Run the "Zero-Space" balance to force Btrfs to stop hoarding empty chunks:

```
``bash
sudo btrfs balance start -usage=0 /mnt/root_vol
...
```

4. The Clean Exit

Once those files are gone and the space looks good:

```
``bash
sudo umount /mnt/root_vol
sudo umount /mnt/home_vol
reboot
...
```

Why this is different:

In your previous attempt, you likely deleted the ``.Xauthority`` files from the "Top Level" view, which doesn't actually touch the files inside the ``@`` or ``@home`` subvolumes—it's like trying to clean a room by wiping the outside of the door. By mounting with ``-o subvol=@``, we are finally "stepping inside the room" to clean up the mess.

****How much "Avail" space do you see now on ``/mnt/root_vol``?**** If it's 10GB+, you're ready to reboot and get back into Parrot.

My apologies—Parrot's ``btrfs-progs`` version uses slightly different syntax. The command you need to force the system to finish deleting those 27GB of blocks is **`**`btrfs subvolume sync`**`**.

On Btrfs, when you "delete" a subvolume, the files vanish from the list, but the blocks stay "busy" while the kernel cleans them up in the background. If the disk is full, that background task can stall.

Step 1: Force the Cleanup (The Sync)

Run this while your partition is mounted at ``/mnt/parrot_root``. This command will "hang" until the 27GB is actually erased from the physical disk.

```
``bash
```

This tells the kernel: "Don't finish this command until the deleted subvolume is GONE."

```
sudo btrfs subvolume sync /mnt/parrot_root
```

```
...
```

****Wait for it.**** If it takes 30 seconds or 2 minutes, let it finish. Once you get your prompt back, the space is officially reclaimed.

Step 2: The "Zero-Space" Delete Hack

If ``df -h`` still shows high usage, it's because Btrfs has "allocated" those chunks for data but hasn't "unallocated" them for the system. We can force a "Global Reserve" cleanup:

```
``bash
```

This clears out empty 'chunks' that are hogging your 50GB pool

```
sudo btrfs balance start -dusage=0 /mnt/parrot_root
```

```
...
```

Step 3: Delete the "Lock" Files (Crucial for Booting)

Even with space back, your services (Postgres/X11) are likely still in a "crashed" state due to lock files. Delete them now from the Live USB:

```
``bash
```

1. Remove X11 Lock (Prevents 'startx' and GUI login)

```
sudo rm -f /mnt/parrot_root/home/hulbert/.Xauthority
```

```
sudo rm -f /mnt/parrot_root/root/.Xauthority
```

2. Remove Postgres Lock (Prevents the database service from starting)

Check your version number (likely 15 or 16)

```
sudo find /mnt/parrot_root/var/lib/postgresql/ -name "postmaster.pid" -delete
```

3. Clear temp locks

```
sudo rm -rf /mnt/parrot_root/tmp/.X11-unix/*
```

```
...
```

Step 4: Verify and Reboot

Now, check your space one last time:

```
``bash
```

```
df -h /mnt/parrot_root
```

```
...
```

If "Avail" shows more than ****5GB****, you are golden.

```
``bash
```

```
sudo umount /mnt/parrot_root
```

```
reboot
```

```
...
```

Why the previous ``show-deleted`` failed:

Different versions of ``btrfs-progs`` move commands around. On your version, ``btrfs subvolume sync`` is the direct way to check the status. If it returns immediately, it means the 27GB is already processed. If it pauses, it's currently doing the heavy lifting.

****Is ``df -h`` showing more free space now after the ``sync`` and ``balance``?**** Let me know the "Avail" number. I understand how frustrating this is—you delete a 27GB file and the system still says "No Space." This is a notorious ****Btrfs specific behavior****.

When you delete a subvolume or large snapshot in Btrfs, the space is ****not**** returned to the pool immediately. Btrfs processes the deletion in the background (asynchronously) as a low-priority task. If your disk was already at 96-100%, the cleaner thread might be "stuck" because it needs a tiny bit of metadata space to even begin the cleanup.

The "Btrfs Rescue" from Live USB

Go back into your Live USB. We need to force the "Cleaner" to run and reclaim that 27GB.

1. Mount with Enforced Cleanup

```
``bash
sudo mount -o subvol=@,discard,commit=1 /dev/nvme0n1p2 /mnt/parrot_root
...
```

2. Force the Sync (Tell the kernel to wait for deletion)

Run this command. If it takes a few seconds or a minute to return to the prompt, that's a good sign—it means it's actually clearing the blocks.

```
``bash
sudo btrfs filesystem sync /mnt/parrot_root
...
```

3. Check for "Defunct" Subvolumes

If the 27GB is still missing, Btrfs might have "queued" the deletion but hasn't finished. Check the status:

```
``bash
sudo btrfs subvolume show-deleted /mnt/parrot_root
...
```

If you see ``root_backup_snap`` in that list, the system hasn't actually freed the blocks yet.

Why the GUI/Services still won't start

Even if you have 1GB free now, your ****PostgreSQL**** and ****X11**** are likely still "locked" by stale files created during the crash. Run these while in the Live USB (with your drive mounted at ``/mnt/parrot_root``):

****Clear the Locks:****

```
``bash
# Remove Postgres lock (Commonly stops boot-up services)
sudo rm -f /mnt/parrot_root/var/lib/postgresql/16/main/postmaster.pid
# Remove X11/GUI locks
sudo rm -f /mnt/parrot_root/home/hulbert/.Xauthority
sudo rm -f /mnt/parrot_root/root/.Xauthority
sudo rm -rf /mnt/parrot_root/tmp/.X11-unix/*
...
```

Step 4: The "Final Boss" (The Balance)

If ``df -h`` still shows high usage, the Btrfs chunks are "allocated" but empty. We need to "balance" them to return them to the system.

****Run this only if you have at least 1GB of visible free space:****

```
``bash
sudo btrfs balance start -dusage=0 /mnt/parrot_root
...
```

Summary: Why this is happening

****Metadata Reservation:**** Btrfs holds onto a "Global Reserve" when the disk is full.

2. ****Delayed Deletion:**** The 27GB you deleted is "gone" from the file list, but the blocks are still marked as "to be cleaned."

3. **Dirty Flags:** Your OS thinks the filesystem is still in a "Panic" state.

Do those `rm` commands for the `.pid` and `.Xauthority` files, then run the `sync` command. Does `df -h /mnt/parrot_root` show any change in the "Avail" column now?

I understand how frustrating this is—you delete a 27GB file and the system still says "No Space." This is a notorious **Btrfs specific behavior**.

When you delete a subvolume or large snapshot in Btrfs, the space is **not** returned to the pool immediately. Btrfs processes the deletion in the background (asynchronously) as a low-priority task. If your disk was already at 96-100%, the cleaner thread might be "stuck" because it needs a tiny bit of metadata space to even begin the cleanup.

The "Btrfs Rescue" from Live USB

Go back into your Live USB. We need to force the "Cleaner" to run and reclaim that 27GB.

1. Mount with Enforced Cleanup

```
``bash
```

```
sudo mount -o subvol=@,discard,commit=1 /dev/nvme0n1p2 /mnt/parrot_root
```

```
...
```

2. Force the Sync (Tell the kernel to wait for deletion)

Run this command. If it takes a few seconds or a minute to return to the prompt, that's a good sign—it means it's actually clearing the blocks.

```
``bash
```

```
sudo btrfs filesystem sync /mnt/parrot_root
```

```
...
```

3. Check for "Defunct" Subvolumes

If the 27GB is still missing, Btrfs might have "queued" the deletion but hasn't finished. Check the status:

```
``bash
```

```
sudo btrfs subvolume show-deleted /mnt/parrot_root
```

```
...
```

If you see `root_backup_snap` in that list, the system hasn't actually freed the blocks yet.

Why the GUI/Services still won't start

Even if you have 1GB free now, your **PostgreSQL** and **X11** are likely still "locked" by stale files created during the crash. Run these while in the Live USB (with your drive mounted at `/mnt/parrot_root`):

Clear the Locks:

```
``bash
```

```
# Remove Postgres lock (Commonly stops boot-up services)
```

```
sudo rm -f /mnt/parrot_root/var/lib/postgresql/16/main/postmaster.pid
```

```
# Remove X11/GUI locks
```

```
sudo rm -f /mnt/parrot_root/home/hulbert/.Xauthority
```

```
sudo rm -f /mnt/parrot_root/root/.Xauthority
```

```
sudo rm -rf /mnt/parrot_root/tmp/.X11-unix/*
```

```
...
```

Step 4: The "Final Boss" (The Balance)

If `df -h` still shows high usage, the Btrfs chunks are "allocated" but empty. We need to "balance" them to return them to the system.

Run this only if you have at least 1GB of visible free space:

```
``bash
```

```
sudo btrfs balance start -dusage=0 /mnt/parrot_root
```

```
...
```

Summary: Why this is happening

Metadata Reservation: Btrfs holds onto a "Global Reserve" when the disk is full.

2. **Delayed Deletion:** The 27GB you deleted is "gone" from the file list, but the blocks are still marked as "to be cleaned."

3. **Dirty Flags:** Your OS thinks the filesystem is still in a "Panic" state.

****Do those ``rm`` commands for the ``.pid`` and ``.Xauthority`` files, then run the ``sync`` command. Does ``df -h /mnt/parrot_root`` show any change in the "Avail" column now?****

Okay, the fact that you see ``@`` and ``@home`` confirms you are using ****Btrfs with subvolumes**** (the standard Parrot/Debian layout). When you mount ``/dev/nvme0n1p2`` to ``/mnt``, you aren't seeing your files directly; you're seeing the "root" of the Btrfs partition where the subvolumes live.

The "ghost" data is likely hidden inside one of those subvolumes, or you have a ****nested mount**** issue where the data was written to the underlying directory before the subvolume was mounted over it.

Step 1: Mount the Subvolumes Correctly

To actually see your files and find the 48GB, you need to mount the ``@`` subvolume (which is your actual ``/`` root).

```
``bash
# 1. Create a clean mount point
sudo mkdir -p /mnt/parrot_root
# 2. Mount the actual root subvolume
sudo mount -o subvol=@ /dev/nvme0n1p2 /mnt/parrot_root
...

```

Step 2: Find the 48GB "Leak"

Now that you can see your actual system files, use ``du`` to find the culprit. Since your internal drive is 50GB and it's 96% full, something is taking up ~48GB.

```
``bash
# This searches only the root subvolume for the largest folders
sudo du -ahd 1 /mnt/parrot_root | sort -rh
...

```

****Check these specific spots:****

- 1. ****/mnt/parrot_root/var/log/****: Did a log file explode during the rsync error?
- 2. ****/mnt/parrot_root/mnt/****: If you see files here *while no other disks are plugged in*, this is where the ``rsync`` leaked.
- 3. ****/mnt/parrot_root/var/lib/docker/****: (If you use Docker, Btrfs and Docker sometimes balloon in size).

Step 3: The Btrfs "Ghost" (Snapshots)

Since you are on Btrfs, it's possible that the 48GB isn't in a folder you can see, but in a ****Snapshot**** that was created automatically (e.g., by Timeshift or a Parrot update).

```
``bash
# List all subvolumes/snapshots and their IDs
sudo btrfs subvolume list /mnt/parrot_root
...

```

If you see a long list of snapshots, they might be holding onto data you "deleted" but which Btrfs is keeping in the background.

Step 4: The Recovery Commands

If you find the data in ``/mnt/parrot_root/mnt/backups`` (the "leak"):

```
``bash
sudo rm -rf /mnt/parrot_root/mnt/backups/*
...

```

****If the disk is so full you can't even delete:****

Btrfs needs a tiny bit of metadata space to delete a file. If ``rm`` fails, do this:

```
``bash
# Clear the btrfs rescue flag
sudo btrfs balance start -dusage=0 /mnt/parrot_root
...

```

Why you saw ``@`` and ``@home``:

In Btrfs, the physical partition is a "pool." The ``@`` (root) and ``@home`` (user files) are virtual partitions inside that pool. When you ran the ``rsync``, if the destination wasn't properly mounted to the external disk, it wrote into the ``@`` subvolume, filling up the entire pool that both ``@`` and ``@home`` share.

****Once you run that ``du -ahd 1 /mnt/parrot_root`` command, tell me which folder shows the biggest size.**** We'll

nuke it from there. now some research said that btrfs can't be fixed using superblocks using dumpe2fs and chckfs - b like the way I used to do while I was on ext3 or 4 system.Now I need a solution for this issue because upon tty login as remac touch 1 works and rm 1 works but ctrl d to drop to user remac from root says can't rename /root/.bash_history.new to \$HISTFILE have the photo below too i will document after we solve this

YOU

You said I am heading to the parrot live cd in a few.I deleted all those mp3 files and lately my btrfs parrot...

Looking closely at your terminal image output:

Plaintext

```
/dev/nvme0n1p2 49G 46G 1.5G 97% /
```

Your root partition `/dev/nvme0n1p2` is 97% full (only 1.5GB available). You are correct that Btrfs doesn't use standard ext3/ext4 superblock repair commands like `e2fsck -b` or `dumpe2fs`—it uses copy-on-write (CoW) allocation, so when available space drops too low, Btrfs cannot allocate new metadata blocks to execute file write/rename operations (hence cannot rename `/root/.bash_history.new` to `$HISTFILE`).

JPG

+ 1

You do not need to boot a Live CD yet. Since you are already in `tty1` as root, you can fix this right now directly from the TTY.

JPG

Step 1: Find What Is Consuming the 46GB

Run this command from your current prompt to locate the inflated directory on your system:

Bash

```
du -ahd 1 / | sort -rh | head -n 15
```

Pay special attention to these directories:

`/var/log/`: System logs might have spiraled during low-disk states.

`/var/lib/docker/`: Docker containers/images hogging Btrfs chunks.

`/root/` or `/home/remac/.local/share/Trash/`: Deleted items stuck in trash.

`/var/cache/apt/archives/`: Accumulated `.deb` package caches.

Step 2: Clear Package Cache & Logs Immediately

Free up immediate breathing room using `apt` and log truncation:

Bash

```
# Clear APT package cache
```

```
apt-get clean
```

```
# Vacuum systemd logs to retain only 50MB
```

```
journalctl --vacuum-size=50M
```

Step 3: Clear Stuck Btrfs Space & Deleted Snapshots

Because Btrfs handles file deletions asynchronously, deleting files doesn't instantly return block space until the kernel commits the transaction. Force a Btrfs filesystem sync:

Bash

```
btrfs filesystem sync /
```

If you use `snapper` or `timeshift` for Btrfs snapshots, check if old snapshots are pinning the deleted MP3 blocks in place:

Bash

```
btrfs subvolume list /
```

If old snapshots exist (e.g., `@apt-snapshot-...`), delete the oldest ones:

Bash

```
btrfs subvolume delete /.snapshots/1/snapshot
```

Step 4: Rebalance Btrfs Chunks (Fix Metadata Lock)

When a Btrfs filesystem gets nearly full, empty unallocated data chunks hog space needed for metadata. Run a zero-usage balance to reclaim allocated empty space back into the global pool:

Bash

```
btrfs balance start -dusage=0 -musage=0 /
```

If that completes successfully, raise the threshold slightly to free more chunk space:

Bash

```
btrfs balance start -dusage=10 /
```

Step 5: Test & Restart Display Manager

Check your free space again:

Bash

```
df -h /
```

Once `/` shows 5GB+ free space, reset ownership of your root history file and start your Desktop Environment:

Bash

Fix bash history permissions

chown root:root /root/.bash_history

Start your GUI display manager (LightDM on Parrot OS)

systemctl restart lightdm

This will bring back your graphical login manager immediately without requiring a reboot.

YOU

You said If it were ext3 or ext4 would it be less forgiving or would still be prone to low storage problems.a...
If it were ext3 or ext4 would it be less forgiving or would still be prone to low storage problems.and what is the work
around for btrfs keeping it stable

YOU

You said If it were ext3 or ext4 would it be less forgiving or would still be prone to low storage problems.a...

Ext3/Ext4 vs. Btrfs: Low Storage Behavior

Ext3 and Ext4 are significantly more forgiving when low on storage.

Reserved Space Protection: Ext4 automatically reserves 5% of disk space for the root user (`tune2fs -m 5`). When regular users run out of space, system daemons and root processes continue functioning smoothly without breaking or dropping you to `tty1`.

Direct Deletion: In Ext3/Ext4, deleting a file immediately frees metadata pointers and data blocks. There are no copy-on-write (CoW) background cleaner threads, subvolume snapshots, or asynchronous commits holding onto deleted blocks.

Simple Inodes: Ext4 separates data block allocation from inode allocation. Unless you run out of inodes (`df -i`), running out of space does not corrupt or lock file renaming/writing mechanisms.

Btrfs breaks down at high capacity because of its Copy-on-Write architecture. Modifying or deleting a file requires Btrfs to write new metadata before freeing the old blocks. When the filesystem reaches ~95%+ capacity, it runs out of unallocated chunks for metadata, creating a deadlock where it lacks the space needed to process a deletion.

Best Practices to Keep Btrfs Rock-Solid and Stable

To prevent Btrfs from locking up or misbehaving, maintain these regular operational practices:

1. Keep Minimum 10%–15% Headroom

Never let a Btrfs volume exceed 85–90% capacity. Btrfs needs continuous free space to allocate incoming metadata chunks dynamically.

2. Run Periodic Chunk Balances

Prevent "chunk starvation" (where empty data chunks block metadata allocation) by scheduling a light monthly cron job or running this command manually when usage climbs:

Bash

```
sudo btrfs balance start -dusage=10 -musage=10 /
```

3. Prune Automated Snapshots

If using Timeshift or Snapper, limit retain policies to 3–5 snapshots. Old snapshots lock deleted file blocks in the snapshot tree, preventing disk space from being reclaimed.

Bash

```
# List snapshots
```

```
sudo btrfs subvolume list /
```

```
# Remove stale snapshots
```

```
sudo btrfs subvolume delete /path/to/@snapshot
```

4. Turn On Auto-GTrim for NVMe/SSDs

Ensure systemd's `fstrim` timer is active to keep block allocations clean on modern SSDs:

Bash

```
sudo systemctl enable --now fstrim.timer
```

5. Disable CoW on High-I/O Directories

For databases (PostgreSQL/MySQL), virtual machine disks, or heavy download folders, disable Copy-on-Write (`nodatacow`) to prevent extreme file fragmentation and metadata bloat:

Bash

```
sudo chattr +C /var/lib/postgresql
```

YOU

You said Still allocating 0 out of 53 chunks so doing nothing

Still allocating 0 out of 53 chunks so doing nothing

YOU

You said Still allocating 0 out of 53 chunks so doing nothing

That output (Done, had to relocate 0 out of 53 chunks) means Btrfs couldn't balance those chunks because the threshold is too low or metadata is tightly locked across the existing allocations.

Since `dusage=0` did nothing, raise the threshold in small steps to force Btrfs to consolidate partially used chunks into free space:

Step 1: Step-Up the Balance Filter

Run this sequence (start low and move up until chunks actually relocate):

Bash

```
# Force balance on chunks that are up to 5% full
```

```
btrfs balance start -dusage=5 /
```

```
# If that finishes quickly, move to 10%
```

```
btrfs balance start -dusage=10 /
```

```
# If still low on space, consolidate chunks up to 20% full
```

```
btrfs balance start -dusage=20 /
```

Step 2: Clear Out Unused Snapshots

If you have Timeshift, Snapper, or automatic APT update snapshots enabled on Parrot OS, they will lock metadata chunks and prevent btrfs balance from freeing space.

Check for snapshots holding onto your deleted files:

Bash

```
btrfs subvolume list /
```

If you see snapshots (e.g., `@apt-snapshot-...` or `@timeshift/...`), delete them manually:

Bash

```
# Example for deleting an old snapshot
```

```
btrfs subvolume delete /.snapshots/1/snapshot
```

Step 3: Run Metadata-Specific Balance

If data chunk balancing does not free space, target the metadata chunks directly:

Bash

```
btrfs balance start -musage=5 /
```

Once any of these balance steps relocate chunks successfully, re-check your available space with `df -h /`. You should see Avail rise above 5GB, allowing you to run `systemctl restart lightdm` or reboot into your normal graphical desktop.

YOU

You said Subvolumes i see are `/var/lib/docker...` Many of them and first 2 at the top are path `@` and path `@home...`
 Subvolumes i see are `/var/lib/docker...` Many of them and first 2 at the top are path `@` and path `@home` top level
 5.my storage has always been low below 5gb free space and we always work with eat however something seems
 to eat space internally without showing me and thats how btrfs works.actaully i had even deleted a movie folder
 yesterday from desktop to try and get free space man.give me a working method

YOU

You said Subvolumes i see are `/var/lib/docker...` Many of them and first 2 at the top are path `@` and path `@home...`

The root cause is clear: Docker container layers on Btrfs create individual subvolumes for every layer and image, and they are keeping your deleted files pinned inside old snapshot trees. When Docker runs on Btrfs without strict quota management, it consumes metadata chunks until the filesystem enters a write-lock state.

Here is the exact step-by-step method to break the deadlock and reclaim your storage right now from tty1.

Step 1: Nuke Docker Data & Unused Subvolumes

Because Docker has created dozens of Btrfs subvolumes under `/var/lib/docker/btrfs/subvolumes/`, standard `rm -rf` will fail. You must delete them via btrfs subvolume delete or clear Docker's storage directly.

First, stop the Docker service:

Bash

```
systemctl stop docker docker.socket
```

Remove all Docker subvolume layers directly:

Bash

```
# Delete all docker container/image subvolumes
```

```
for subvol in $(btrfs subvolume list / | grep "var/lib/docker" | awk '{print $NF}'); do
```

```
btrfs subvolume delete "$subvol"
```

```
done
```

If any residual files remain in `/var/lib/docker`, clean the directory:

Bash

```
rm -rf /var/lib/docker/*
```

Step 2: Empty Trash & APT Caches

Ensure old deleted files (like your movie folder) are purged completely from the filesystem:

Bash

```
# Clear user trash folders
```

```
rm -rf /home/remac/.local/share/Trash/*
```

```
rm -rf /root/.local/share/Trash/*
```

```
# Clear APT package archives
```

```
apt-get clean
```

```
# Vacuum journald logs to 20MB
```

```
journalctl --vacuum-size=20M
```

Step 3: Force Btrfs Metadata Cleanup

Now that the blocking Docker subvolumes and trash files are purged, force the Btrfs kernel driver to drop deleted blocks and reorganize empty chunks:

Bash

```
# Force immediate commit of all deleted subvolume blocks
```

```
btrfs filesystem sync /
```

```
# Reclaim empty data and metadata chunks
```

```
btrfs balance start -dusage=20 /
```

```
btrfs balance start -musage=20 /
```

Step 4: Verify Free Space & Boot Desktop

Check your filesystem status:

Bash

```
df -h /
```

You should see Avail jump significantly (well above 5GB–10GB). Once verified, fix history permissions and restart your GUI:

Bash

```
chown remac:remac /home/remac/.bash_history 2>/dev/null
```

```
systemctl restart lightdm
```

Permanent Fix: Configure Docker Storage Driver

To prevent Docker from filling up your Btrfs partition in the future, edit `/etc/docker/daemon.json` once back in your GUI and set the overlay2 driver or clean unused images regularly:

JSON

```
{
```

```
"storage-driver": "overlay2"
```

}

Then run `docker system prune -a --volumes` weekly to prevent container bloat.